HDOC DAY – 44

MAIN ALGORITHM:

1. Start.
2. Display the menu:
    - Reverse each word in a sentence.
    - Print initials of a name.
    - Exit.
3. Ask the user to enter a choice.
4. Use switch to perform the selected operation.
5. If choice is 1:
    - Read a sentence.
    - Call reverseEachWord().
6. If choice is 2:
    - Read a name.
    - Call printInitials().
7. If choice is 3, terminate the program.
8. For any other choice, display "Invalid choice".
9. Repeat the menu until the user selects Exit.
10. Stop.

Algorithm for reverseEachWord():

1. Start from the first character of the sentence.
2. Find the beginning of a word.
3. Move forward until a space or the end of the string is found.
4. Reverse the characters of that word using two positions:

    - one at the beginning,
    - one at the end.
5. Swap the characters until the two positions meet.
6. Repeat for all words.
7. Display the modified sentence.

Algorithm for printInitials():

1. Print the first character of the name as the first initial.
2. Traverse the name character by character.
3. Whenever a space is found, check the next character.
4. If the next character is not a space or '\0', print it as the next initial.
5. Print . after every initial.
6. Stop.

C PROGRAM:

```
#include <stdio.h>
#include <string.h>

void reverseEachWord(char str[]);
void printInitials(char name[]);

int main()
{
   int choice;
   char str[200];
```

```
do
{
   printf("\n--- MENU ---\n");
   printf("1. Reverse each word in a sentence\n");
   printf("2. Print initials of a name\n");
   printf("3. Exit\n");

   printf("Enter your choice: ");
   scanf("%d", &choice);
   getchar();   // Removes newline left by scanf()

   switch(choice)
   {
      case 1:
         printf("Enter a sentence: ");
         fgets(str, sizeof(str), stdin);

         // Remove newline added by fgets()
         str[strcspn(str, "\n")] = '\0';

         reverseEachWord(str);

         printf("Output: %s\n", str);
         break;
```

```
        case 2:
            printf("Enter a name: ");
            fgets(str, sizeof(str), stdin);

            str[strcspn(str, "\n")] = '\0';

            printf("Initials: ");
            printInitials(str);
            break;

        case 3:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice!\n");
      }

   } while(choice != 3);

   return 0;
}


/* Function to reverse each word */
```

```
void reverseEachWord(char str[])
{
   int start = 0, end, i;
   char temp;

   for(i = 0; ; i++)
   {
      // Space or end of string means a word has ended
      if(str[i] == ' ' || str[i] == '\0')
      {
         end = i - 1;

         // Reverse the current word
         while(start < end)
         {
            temp = str[start];
            str[start] = str[end];
            str[end] = temp;

            start++;
            end--;
         }

         start = i + 1;
      }
```

```
        if(str[i] == '\0')
            break;
    }
}


/* Function to print initials */
void printInitials(char name[])
{
    int i;

    // Print first initial
    if(name[0] != '\0')
        printf("%c.", name[0]);

    for(i = 0; name[i] != '\0'; i++)
    {
        // Character after a space is the next initial
        if(name[i] == ' ' &&
           name[i + 1] != ' ' &&
           name[i + 1] != '\0')
        {
            printf("%c.", name[i + 1]);
        }
```

}

printf("\n");

}

```c
#include <stdio.h>
#include <string.h>

void reverseEachWord(char str[]);
void printInitials(char name[]);

int main()
{
    int choice;
    char str[200];

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();   // Removes newline left by scanf()

        switch(choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(str, sizeof(str), stdin);

                // Remove newline added by fgets()
                str[strcspn(str, "\n")] = '\0';

                reverseEachWord(str);

                printf("Output: %s\n", str);
                break;

            case 2:
                printf("Enter a name: ");
                fgets(str, sizeof(str), stdin);

                str[strcspn(str, "\n")] = '\0';

                printf("Initials: ");
                printInitials(str);
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice!\n");
        }
```

```
Session  Actions  Edit  View  Help
/* Function to reverse each word */
void reverseEachWord(char str[])
{
    int start = 0, end, i;
    char temp;

    for(i = 0; ; i++)
    {
        // Space or end of string means a word has ended
        if(str[i] == ' ' || str[i] == '\0')
        {
            end = i - 1;

            // Reverse the current word
            while(start < end)
            {
                temp = str[start];
                str[start] = str[end];
                str[end] = temp;

                start++;
                end--;
            }

            start = i + 1;
        }

        if(str[i] == '\0')
            break;
    }
}

/* Function to print initials */
void printInitials(char name[])
{
    int i;

    // Print first initial
    if(name[0] != '\0')
        printf("%c.", name[0]);

    for(i = 0; name[i] != '\0'; i++)
    {
        // Character after a space is the next initial
        if(name[i] == ' ' &&
           name[i + 1] != ' ' &&
           name[i + 1] != '\0')
        {
            printf("%c.", name[i + 1]);
        }
    }
```

TEST CASE TABLE:

Test Case	Choice	Input	Expected Output
1	1	What is special?	tahW si ?laiceps
2	1	Hello World	olleH dlroW

3	1	C Programming	C gnimmargorP
4	2	Alexander Grahambell	A.G.
5	2	Ratan Naval Tata	R.N.T.
6	2	John Doe	J.D.
7	3	—	Program terminates
8	5	—	Invalid choice!

